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Capital Budgeting, Uncertainty, and Misallocation

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1 Big picture

- **Question.** How much do capital budgeting practices reduce the productivity and firm-value losses created by uncertainty and time-to-build frictions?
- **Standard friction.** In canonical time-to-build investment models, firms must commit investment before all relevant information arrives. Forecast mistakes then produce overinvestment or underinvestment and reduce aggregate productivity.
- **Paper's key idea.** Real firms engage in capital budgeting. They can acquire information before setting plans and can adjust investment after plans are set. These two margins reduce misallocation caused by uncertainty.
- **Two mechanisms.** The paper studies (i) ex-ante information acquisition, which improves forecast precision, and (ii) ex-post investment flexibility, which lets actual investment deviate from initial plans once new information arrives.
- **Data innovation.** The empirical analysis uses Japanese Ministry of Finance firm-level survey data containing firms' internal forecasts of sales and profits, investment plans, and realized investment expenditures.
- **Main empirical facts.** Firms face sizable uncertainty when making investment plans; higher-productivity firms face lower uncertainty; investment plans are flexible; and realized investment responds to new information about productivity.
- **Model contribution.** The authors embed information acquisition and state-contingent investment planning into a neoclassical investment model.
- **Theoretical result.** The marginal benefit of learning is increasing in firm productivity. This makes information and productivity complements.
- **Calibration result.** State-contingent investment flexibility reduces aggregate productivity losses from uncertainty by about 41%, while information allocation reduces losses by about 17%.
- **Distributional result.** Flexibility mainly benefits lower-productivity firms; information acquisition mainly benefits higher-productivity firms.
- **Economic takeaway.** Capital budgeting is not just internal accounting. It is a technology for reducing uncertainty-driven capital misallocation.

2 Data and measurement

2.1 Firm panel

- The sample is drawn from Japanese Ministry of Finance data covering manufacturing firms over fiscal years 2005–2016.
- The dataset contains firm-year financial statements plus management forecasts and plans.
- The main variables are realized sales and profits, expected sales and profits, planned capital investment, and realized capital investment.
- The final sample contains 2,040 firms and a mix of public and private firms.
- Firms with zero sales or missing labor-cost and PPE information are excluded.

Interpretation: The survey is especially useful because it observes both expectations and realizations. This allows the paper to separate ex-ante planning, forecast errors, and ex-post deviations from the plan.

2.2 Investment plans and investment flexibility

Planned capital expenditures are interpreted as pre-committed capital purchases at the beginning of the fiscal year. Realized capital investment is measured at the end of the fiscal year.

The paper defines investment plan deviation as

$$\log k_{i,t+1} - \log k_{i,t+1}^P,$$

where k^P is planned next-period capital and k is realized next-period capital.

Interpretation: If investment is fully inflexible, realized investment should equal planned investment. Dispersion in plan deviations is evidence that firms can respond to new information after plans are made.

2.3 Revenue productivity and TFP forecast errors

The paper computes a revenue-productivity measure using a Cobb-Douglas value-added production function. A simplified expression for productivity is

$$\tilde{z} = \frac{(py)^{\eta/(\eta-1)}}{k^\alpha l^{1-\alpha}},$$

where py is value added, k is capital, l is labor, η is demand elasticity, and α is the capital intensity parameter.

Forecast errors are computed using expected sales:

$$\hat{\varepsilon}^{FE} = \log \frac{(py)^{\eta/(\eta-1)}}{k^\alpha l^{1-\alpha}} - \log \frac{(py^e)^{\eta/(\eta-1)}}{k^\alpha l^{1-\alpha}},$$

where py^e is forecasted value added.

Interpretation: The forecast error measures what firms did not know when they set investment plans. It is the central empirical proxy for uncertainty around productivity.

2.4 Firm uncertainty

The paper distinguishes two uncertainty concepts:

- **forecast error level:** the realized difference between forecasted and actual productivity;
- **residual uncertainty:** dispersion of forecast errors after conditioning on prior productivity bins.

Interpretation: A firm can be uncertain even if a single realized forecast error is small. The dispersion of possible forecast errors is the uncertainty the firm faces while planning.

2.5 Investment flexibility measurement

The key empirical object is the gap between actual and planned investment. The paper reports the distribution of

$$\frac{i - i^P}{k_{t-1}},$$

where i is realized investment, i^P is planned investment, and k_{t-1} is beginning-of-period capital.

Interpretation: Large deviations from plans indicate that firms revise investment after learning new information. The model interprets this as costly but valuable investment flexibility.

3 Empirical specifications and stylized facts

3.1 Uncertainty and productivity

The first regression relates forecast errors to lagged productivity:

$$\log(1 + |\widehat{\varepsilon}_{j,t}^{FE}|) = \alpha_j + \alpha_t + \beta \log \tilde{z}_{j,t-1} + \eta_{j,t}.$$

Interpretation: A negative β means high-productivity firms face less forecast uncertainty. The paper finds this relationship both across firms and within firms over time.

3.2 Planned investment and expected productivity

The second regression studies whether planned investment reflects expected productivity:

$$\left(\frac{i^p}{k}\right)_{j,t} = \alpha_{j,t} + \beta_1 E[\log \tilde{z}_{j,t}] + \eta_{j,t}.$$

Interpretation: If investment plans are meaningful and forward looking, planned investment should rise with expected productivity.

3.3 Realized investment and new information

The third regression asks whether realized investment responds to realized productivity after controlling for plans:

$$\left(\frac{i}{k}\right)_{j,t} = \alpha_{j,t} + \beta_1 \log \tilde{z}_{j,t} + \beta_2 \left(\frac{i^p}{k}\right)_{j,t} + \eta_{j,t}.$$

Interpretation: If actual investment is flexible, realized investment should respond to new productivity information even after controlling for planned investment.

3.4 Main empirical facts

- Forecast errors are dispersed: investment plans are made under meaningful uncertainty.
- Residual uncertainty varies across firms.
- Higher-productivity firms tend to face lower uncertainty.
- Actual investment deviates substantially from investment plans.
- Planned investment is positively related to expected productivity.
- Realized investment is positively related to realized productivity even after controlling for planned investment.

Interpretation: These facts jointly motivate a model where firms both learn before planning and adjust after planning.

4 Model and theoretical mechanisms

4.1 Economic environment

The model is an infinite-horizon economy with heterogeneous intermediate-goods firms. Each period has three subperiods:

1. **information acquisition:** the manager chooses signal precision before observing current productivity;

2. **investment plan formation:** the manager makes an investment plan based on the signal;
3. **investment implementation:** after actual productivity is observed, the manager chooses realized investment subject to a deviation cost.

Interpretation: The three-subperiod structure maps directly into the data: expectations and plans are measured early, and realizations are measured later.

4.2 Productivity process and signal

Persistent productivity follows

$$\log z_{i,t} = \rho \log z_{i,t-1} + \varepsilon_{i,t}, \quad \varepsilon_{i,t} \sim N(0, \sigma_\varepsilon^2).$$

The manager chooses a signal precision and observes

$$s_{i,t} = \mu_{i,t} + \varepsilon_{i,t} + u_{i,t}, \quad u_{i,t} \sim N(0, \sigma_{i,t}^2).$$

Interpretation: Better information means a lower posterior variance about current productivity. More precision is valuable but costly.

4.3 Cost of information acquisition

The cost of reducing uncertainty is specified as

$$C(\sigma_{i,t}, z_{i,t-1}, \mu_i) = \xi \mu_i^{\zeta_\mu} z_{i,t-1}^{\zeta_z} \sigma_{i,t}^{-2}.$$

Interpretation: The parameters ζ_μ and ζ_z allow the cost of learning to vary with permanent productivity and lagged productivity. This is crucial for matching heterogeneity in uncertainty.

4.4 Investment plan deviation cost

After actual productivity is observed, the firm may deviate from its plan. The cost is

$$\phi(k_{i,t+1}^p, k_{i,t+1}) = \frac{\chi}{2} \left(\frac{k_{i,t+1}}{k_{i,t+1}^p} - 1 \right)^2 k_{i,t+1}^p.$$

Interpretation: The parameter χ governs flexibility. A high χ means it is costly to revise plans; $\chi = \infty$ corresponds to complete inflexibility.

4.5 Bellman equations

Let $J^\mu(z, k, k^p)$ be the value after actual productivity is observed:

$$J^\mu(z, k, k^p) = \max_{k'} \left[\pi(z, k, k') - (1 - \delta)k - k' - \frac{\chi}{2} \left(\frac{k'}{k^p} - 1 \right)^2 k^p + \frac{1}{1 + r} V^\mu(k', z) \right].$$

After observing the signal, the manager chooses an investment plan:

$$W^\mu(z_{-1}, s, k, \sigma) = \max_{k^p} E[J^\mu(z, k, k^p) \mid z_{-1}, s, \sigma].$$

Before the signal is observed, the manager chooses signal precision:

$$V^\mu(k, z_{-1}) = \max_{\sigma} [-C(\sigma, z_{-1}, \mu) + E[W^\mu(z_{-1}, s, k, \sigma) \mid z_{-1}]].$$

Interpretation: The nested timing allows the paper to quantify the separate values of learning before planning and adjusting after planning.

4.6 Proposition 1: marginal benefit of learning

In the simplified case with $\chi = \infty$, the ex-ante firm value is decreasing in posterior uncertainty, and the value of reducing posterior uncertainty is increasing in productivity.

$$MB = \frac{1}{1+r} \frac{\partial E[V(\cdot)]}{\partial \mathbb{V}^{-1}} = \frac{1}{1+r} \bar{h}(\mu z_{-1}^{\rho})^{\Theta_z/(1-\Theta_k)} \frac{\partial \mathcal{F}^{\pi}(\mathbb{V}^{-1})}{\partial \mathbb{V}^{-1}}.$$

Interpretation: Information is an intangible production input. The return to information rises with productivity, so productive firms endogenously want more precise information.

4.7 Proposition 2: information allocation and aggregate productivity

For any fixed marginal distributions of lagged productivity, permanent productivity, and information choices, aggregate TFP falls when information choices are less correlated with productivity.

Interpretation: This formalizes information misallocation. Aggregate productivity is higher when information is allocated toward more productive firms.

5 Calibration and counterfactuals

5.1 Calibration strategy

- Some parameters are estimated externally from the data: demand elasticity, capital share, productivity persistence, volatility, and dispersion of fixed effects.
- Other parameters are calibrated internally: investment flexibility χ , learning cost level ξ , and cost heterogeneity parameters ζ_z and ζ_{μ} .
- The calibration targets moments related to forecast errors, investment plan deviations, and correlations of productivity with uncertainty.

Interpretation: The model is disciplined by both static moments and dynamic moments. This lets the authors separately identify investment flexibility and information acquisition costs.

5.2 Value of investment flexibility

The paper compares the baseline model with a counterfactual in which investment plans cannot be adjusted ex post. Firm value is computed as

$$\Pi(\mu, z_{-1}) = E_{z_{-1}} \sum_{t=1}^{\infty} \left(\frac{1}{1+r} \right)^t [\pi(\mu, z_t, k_t) - (r + \delta)k_t].$$

The value gain from flexibility is

$$\Delta \Pi(\mu, z_{-1}) = 100 \times \left(\frac{\Pi^{baseline}(\mu, z_{-1})}{\Pi^{counterfactual}(\mu, z_{-1})} - 1 \right).$$

Interpretation: Flexibility is valuable because it lets firms revise investment after realizing productivity. The gains are larger for lower-productivity firms in the calibrated model.

5.3 Value of information allocation

The second counterfactual makes information allocation homogeneous across firms, removing the skew toward high-productivity firms. The authors then compute firm-value and aggregate-productivity gains from restoring the baseline allocation.

Interpretation: Information acquisition is most valuable for higher-productivity firms because information and productivity are complements. Reallocating information toward these firms raises aggregate productivity.

5.4 Aggregate results

- Baseline uncertainty generates TFP losses of about 0.20%.
- Removing investment flexibility increases losses to about 0.34%.
- Homogeneous information allocation generates losses of about 0.24%.
- Therefore investment flexibility reduces aggregate productivity losses by roughly 41%.
- Information allocation reduces losses by roughly 17%.

Interpretation: Flexibility is the quantitatively larger mechanism in the calibration, but information allocation also matters because it directs intangible information resources toward productive firms.

6 Tables and figures

This section keeps the compact visual layout used in the previous notes. Adjacent figures and tables from the original paper are grouped together where possible.

6.1 Table 1 + Figure 1: sample and uncertainty distribution

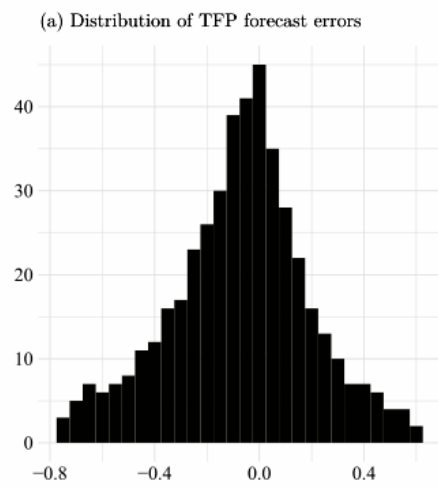
Table 1

Summary statistics. The table shows the summary statistics of our firm-year panel. Our sample of interest is the manufacturing sector, defined in accordance with the Japanese Standard Industrial Classification. The total number of firms in our sample is 2,040, of which 46% are publicly listed and the rest are private companies. Capital stock is the Net Plants, Property & Equipment. Profits are "ordinary profits" according to Japanese Generally-Accepted Accounting Principles (GAAP). The Wage Bill is the sum of total salary cost for employees and company officers as well as the bonus for employees and company officers. Employment is the number of full-time equivalent workers and may include fractions. All monetary values are reported in current yen and we winsorize all variables at the 1% level.

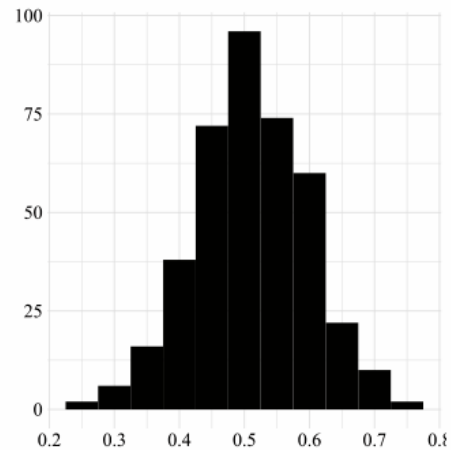
Variable	Mean	SD	Percentile		
			25 th	50 th	75 th
Total Assets (mn)	87,223.8	368,988.8	7,957.5	20,554.0	56,933.0
Capital Stock (mn)	46,816.1	225,075.7	2,840.2	8,579.0	26,003.2
Employment (count, FTE)	1,052.0	2,872.6	190.0	428.0	946.0
COGS (mn)	65,209.1	290,962.3	5,847.0	15,267.0	40,751.0
Wage Bill (mn)	7,501.9	26,098.2	989.0	2,485.0	5,935.0
Sales (mn)	99,450.9	390,199.6	12,389.2	27,096.0	68,015.0
Expected Sales (mn)	100,556.2	386,181.3	12,889.5	27,700	69,000
Profits (mn)	5,585.2	43,170.5	223.0	1062.0	3,521.0
Expected Profits (mn)	5,493.50	37,757.00	300	1,070	3,400
Total Capital Investment (mn)	3,909.4	14,365.9	273.0	821.5	2,533.8
Total Capital Investment Plan (mn)	4,347.10	15,627.40	300	900	2,730.50
Investment (% Capital)	10.1	44.2	3.0	6.0	11.4
Capital Investment Plan (% Capital)	12.3	62.8	3.0	6.6	12.8

(a) Summary statistics.

(a) Distribution of TFP forecast errors



(b) Distribution of the dispersion of scaled TFP forecast errors within TFP centiles



1. The distribution of firm-level uncertainty in Japan. Fig. 1a plot

(b) Distribution of TFP forecast errors and residual uncertainty.

Read:

- Table 1 reports the firm-year summary statistics for the Japanese manufacturing panel.
- Firms in the sample are large: average total assets are about 87,224 million yen.
- Planned capital investment averages about 4,347 million yen, and the median investment plan is about 6.6% of capital.
- Figure 1 shows a dispersed distribution of TFP forecast errors.
- It also shows substantial heterogeneity in residual uncertainty across TFP percentiles.

Economic message: The data directly observe both ex-ante plans and ex-post realizations. This makes it possible to measure uncertainty and flexibility rather than infer them indirectly.

6.2 Table 2 + Table 3: uncertainty facts

Table 2

Statistics of distributions of TFP and investment plan deviations. The table reports statistics associated with the distributions reported in Figs. 1 and 2. In the first row, TFP forecast errors have been deflated by a factor of $1 - (1 - \alpha) \frac{1-\alpha}{\eta}$ and normalized by σ_e . In the second row, standard deviation of TFP forecast errors is normalized by the volatility of TFP within the same centile of TFP. In the third and fourth rows, we report investment plan deviations, where the fourth row excludes zero planned investment. K refers to initial period capital stock. Both are reported in decimals. We winsorize all variables at the 1% level.

Variable	Mean	SD	Percentile		
			25 th	50 th	75 th
TFP Forecast Error	-0.099	0.488	-0.361	-0.079	0.152
SD of TFP Forecast Errors $ z_{j,t-1} $ (fraction of $\sigma_e z_{j,t-1} $)	0.509	0.085	0.446	0.513	0.568
Investment Plan Deviations (fraction of K)	-0.003	0.062	-0.024	-0.005	0.01
Investment Plan Deviations (fraction of K , excl. $i^P = 0$)	-0.008	0.061	-0.025	-0.006	0.009

(a) Forecast-error and investment-plan-deviation statistics.

Table 3

Uncertainty and correlation with TFP. The table reports our estimation of $\log \left(1 + |\widehat{\varepsilon}_{j,t}^{FE}| \right) = \alpha_{j,t} + \beta \log \tilde{z}_{j,t-1} + \eta_{j,t}$. Columns 1 and 2 report the OLS specifications while Columns 3 and 4 report Poisson regression specifications. $\widehat{\varepsilon}_{it}^{FE}$ refers to TFP forecast errors and $\log \tilde{z}_{j,t-1}$ refers to realized productivity at $t - 1$. Observations are at the firm-year level. All variables are winsorized at the 1% level. Fixed effects used across specifications as indicated. Standard errors are presented in parentheses and clustered by firm.

	(1)	(2)	(3)	(4)
	$\log(1 + \widehat{\varepsilon}_{it}^{FE})$		$ \widehat{\varepsilon}_{it}^{FE} $	
$\log \tilde{z}_{j,t-1}$	-0.012*** (0.003)	-0.021*** (0.007)	-0.060*** (0.016)	-0.105*** (0.029)
Method	OLS	OLS	Poisson	Poisson
Industry FE	Y	Subsumed	Y	Subsumed
Firm FE	N	Y	N	Y
Year FE	Y	Y	Y	Y
Observations	15,230	15,230	15,230	15,230
R^2	0.180	0.465	0.033	0.079

(b) Uncertainty and correlation with productivity.

Read:

- Table 2 reports that normalized forecast errors are large, with a standard deviation around 0.488.
- Residual forecast uncertainty has a mean around 0.509 and an interquartile range roughly from 0.446 to 0.568.
- Investment plan deviations are small on average but have meaningful dispersion.
- Table 3 shows that forecast-error magnitude is negatively related to prior productivity.
- The relationship is stronger when firm fixed effects are included, indicating within-firm changes in productivity are important.

Economic message: Firms face uncertainty when planning, and more productive firms tend to face lower uncertainty. This motivates endogenous information acquisition.

6.3 Figure 2 + Table 4: investment flexibility

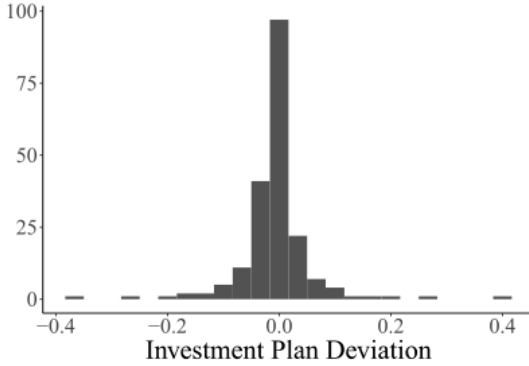


Fig. 2. Distribution of investment plan deviations. Fig. 2 plots the histogram of investment plan deviations, defined as $\frac{i - p}{k}$, for the full sample of manu

(a) Distribution of investment plan deviations.

Table 4

The correlation of expected and realized investment with productivity. Columns 1 and 2 below report our estimation of $\left(\frac{i}{k}\right)_{j,t} = \alpha_{j,t} + \beta_1 E[(\log \tilde{z}_{j,t})_{j,t}] + \eta_{j,t}$ with industry, year and firm, year fixed effects, respectively. Columns 3 and 4 below report our estimation of $\left(\frac{i}{k}\right)_{j,t} = \alpha_{j,t} + \beta_1 \log \tilde{z}_{j,t} + \beta_2 \left(\frac{i}{k}\right)_{j,t} + \eta_{j,t}$. Column 2 reports a linear model that omits investment plans, while Column 3 includes investment plans. Observations are at the firm-year level. $\left(\frac{i}{k}\right)_{j,t}$ and $\left(\frac{i}{k}\right)_{j,t}$ refer to realized and planned investment. $\log \tilde{z}_{j,t}$ refers to realized productivity using our baseline method. All variables are winsorized at the 1% level. All specifications include firm and time fixed effects. Standard errors are presented in parentheses and clustered by firm.

	(1)	(2)	(3)	(4)
	$\left(\frac{i}{k}\right)_{j,t}$		$\left(\frac{i}{k}\right)_{j,t}$	
$E[\log \tilde{z}_{j,t}]$	2.399*** (0.234)	2.301*** (0.279)		
$\log \tilde{z}_{j,t}$			4.245*** (0.452)	2.085*** (0.319)
$\left(\frac{i}{k}\right)_{j,t}$				0.633*** (0.018)
Industry FE	Y	Subsumed		
Firm FE		Y	Y	Y
Year FE	Y	Y	Y	Y
Observations	18,624	18,624	17,926	14,959
R^2	0.055	0.687	0.545	0.752

(b) Expected and realized investment with productivity.

Read:

- Figure 2 shows a concentrated but non-degenerate distribution of investment plan deviations.
- Many firms revise investment downward, consistent with partly flexible plans.
- Table 4 shows planned investment is positively related to expected productivity.
- Realized investment is positively related to realized productivity even after controlling for investment plans.
- The coefficient on realized productivity is larger when plans are omitted, showing that plans already embed some productivity information.

Economic message: Investment plans are neither arbitrary nor perfectly rigid. They incorporate forecasts, but actual investment still responds to new information.

6.4 Figure 3 + Proposition 1: model timing and learning

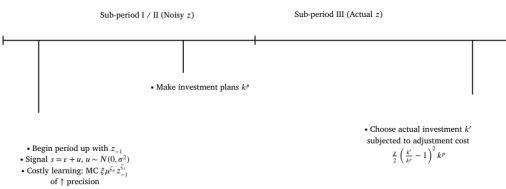


Fig. 3. Model timing. This figure illustrates the three sub-periods of our model.

(a) Model timing.

friction. As such, the impact of information frictions must be decreasing in investment flexibility. In the latter sections, our quantitative model will allow us to quantify the value of investment flexibility accounting for such information frictions.

However, it is unclear how the information frictions affect the value of information on net, holding fixed investment flexibility. Specifically, we are interested in using our model to rationalize our empirical observation that there is a negative correlation between productivity and uncertainty in the data, as well as to why the correlation of productivity and uncertainty is stronger within firm. As such, for the rest of this subsection, we will focus on presenting a characterization of the value of information for firms and the aggregate economy with the simplifying assumption that $\gamma \rightarrow \infty$. We then build on these insights to examine their quantitative implications in Section 5. Appendix C.3 shows that these same insights apply to our more general model with investment flexibility.

3.5.1. What drives heterogeneity in information acquisition?

To begin, we first state the following formal proposition, which will

functions of information choices because the investment decision is tied back to the information set of the firm during the capital budgeting process. Thus, one can interpret information as an intangible asset and price the value of a firm in terms of its sequence of information choices.

Moreover, this reveals that even if firms had access to the same technology for information acquisition, we would already see an endogenous distribution of uncertainty because firms invest differentially into information acquisition based on their productivity. Therefore, this allows us to interpret heterogeneity in uncertainty as a form of factor allocation, which we term “information allocation”.

It is important to point out that the impact of μ and z_{t-1} on the value of information derives from fundamentally different mechanisms. In the case of μ , this has the standard interpretation that permanently higher productivity firms have higher incentives to invest in more assets, and are thus larger firms in the longer run.

In contrast, the correlation of the marginal benefit with z_{t-1} captures how firm incentives fluctuate with idiosyncratic shocks to firm productivity. In turn, this reflects the incentives of a firm to dynamically reallocate resources dedicated to information acquisition, invest-

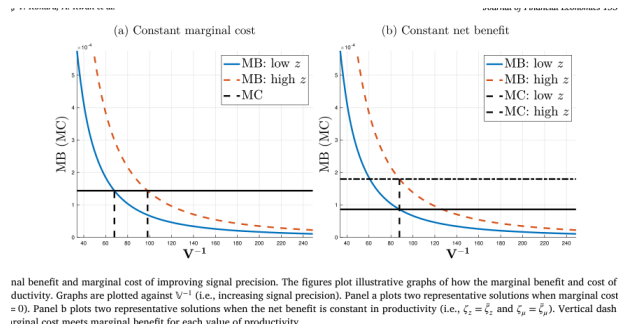
(b) Proposition 1 and marginal benefit expression.

Read:

- Figure 3 separates information acquisition, investment plan formation, and actual investment.
- The model maps directly to the data: forecasts and plans are observed before realized productivity and investment.
- Proposition 1 states that firm value rises as posterior uncertainty falls.
- It also states that the value of reducing uncertainty increases with productivity.

Economic message: The model formalizes capital budgeting as a sequence of learning, planning, and state-contingent investment adjustment.

6.5 Figure 4 + Proposition 2: information as an input



(a) Marginal benefit and marginal cost of learning.

Proposition 2: For any given marginal distribution of initial period productivity z_{-1} , long-run productivity μ , and information choice V , aggregate TFP is decreasing in the correlation between productivity and V .

The result is analogous to a model where the distribution of physical factor inputs is arbitrarily imposed, and the total stock of factors is fixed such that TFP gains arise solely from factor reallocation. In the context of physical factors, because productivity and factors of production are explicitly modeled as complements it is obvious that TFP is decreasing in the correlation between productivity and V .

Read:

- Figure 4 plots marginal benefits and marginal costs of improving signal precision.
- High-productivity firms have higher marginal benefits of learning.
- The optimal precision is found where marginal benefit equals marginal cost.
- Proposition 2 states that aggregate TFP falls when information is less positively correlated with productivity.

Economic message: Information acts like an intangible factor of production. Allocating information to high-productivity firms raises aggregate productivity.

6.6 Table 5 + Table 6: calibration

Table 5

Calibrated parameters. The table reports our parameter values taken directly from our data (externally estimated), fixed parameters, or internally calibrated. Externally estimated parameters follow the methodology outlined in Appendix D.1.

Parameter Description		Values
<u>Externally Estimated Parameters</u>		
η	Elasticity of substitution	4.682
α	Capital share	0.360
ρ	Autocorrel. of physical TFP	0.426
σ_ε	Volatility of physical TFP	0.246
σ_μ	Dispersion of fixed effects	0.546
<u>Fixed Parameters</u>		
r	Interest rate	0.02
δ	Depreciation rate	0.06
N	Labor supply	1
P	Price of output good	1
<u>Internally Estimated Parameters</u>		
χ	Flexibility of investment	0.110
ξ	Average marginal cost	1.678×10^{-5}
ζ_z	Correlation of marginal cost with z	0.255
ζ_μ	Correlation of marginal cost with μ	3.460

(a) Calibrated parameters.

Table 6

targeted moments. The table reports the moments targeted in the calibration routine.

Moments		
$\frac{sd(\varepsilon^{FE})}{\sigma_\varepsilon}$	Ratio of TFP Forecast Error to TFP Shock	1.342
$\sigma\left(\frac{\xi}{k^p}\right)$	Volatility of Investment Plan Deviation	0.043
$\rho(z_{-1}, \varepsilon)$	Corr. of Past TFP with TFP Forecast Error	-0.021
$\rho(z_{-1}, \varepsilon)$ without firm FE	Corr. of Past TFP with TFP Forecast Error, no Firm FE	-0.012

(b) Targeted moments.

Read:

- Table 5 reports externally estimated parameters and internally calibrated parameters.
- The estimated flexibility parameter is $\chi = 0.110$.
- The cost of information acquisition is governed by ξ , ζ_z , and ζ_μ .
- Table 6 reports calibration targets: forecast-error dispersion, plan-deviation dispersion, and correlations between productivity and uncertainty.

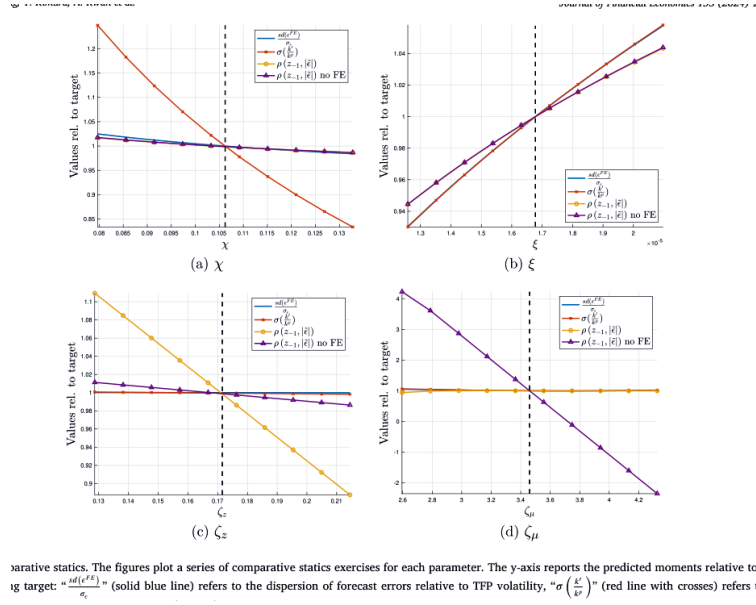
Economic message: The calibration matches moments that separately discipline learning costs and investment flexibility.

6.7 Figure 5: comparative statics

Read:

- Increasing χ mainly changes the standard deviation of investment plan deviations.
- Increasing ξ affects both forecast-error dispersion and plan-deviation dispersion.
- ζ_z is most related to within-firm correlations of productivity with forecast errors.
- ζ_μ is most related to cross-firm correlations.

Economic message: The comparative statics explain why the targeted moments identify different parts of the capital-budgeting technology.



6.8 Figure 6 + Tables 7–8: counterfactual values

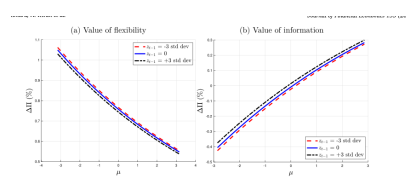


Figure 7: The impact of investment flexibility and information allocation on firm value. The figures plot ΔV against μ for three values of z_{-1} . The range of μ covers three standard deviations of z_{-1} around the mean. Solid blue line plots ΔV for $z_{-1} = 0$, dashed red line for $z_{-1} = 2$ standard deviations right of the mean, and dotted black line for $z_{-1} = 3$ standard deviations right of the mean. Fig. 7a vs 7b

(a) Private value of flexibility and information.

Table 7
The impact of investment flexibility and information allocation on firm value. Panel A reports distributional statistics associated with $\Delta V(\mu, z_{-1})$ for the counterfactual experiment quantifying the value of investment flexibility, while Panel B reports the same distributional statistics for the counterfactual experiment quantifying the value of information allocation. Probabilities might not sum to 1 due to rounding. Column (5) reports the change in average firm value, while Column (6) reports the change in aggregate corporate value (Eq. (16)). The coefficients in Columns (7) and (8) refer to changes in profit attributed to the equation: $\Delta \Pi = \beta_1 + \beta_2 \log z_{-1} + \beta_3 \log \mu$.

Panel A: Investment Flexibility							
	$P(\Delta V < 0)$	$P(\Delta V > 0)$	$E(\Delta V \Delta V < 0)$	$E(\Delta V \Delta V > 0)$	Δ / Π	β_1	β_2
PE	0	1	—	0.759	1.133	-0.016	-0.149
GE	0	1	—	0.272	0.137	-0.016	-0.148
Panel B: Information Allocation							
	$P(\Delta I < 0)$	$P(\Delta I > 0)$	$E(\Delta I \Delta I < 0)$	$E(\Delta I \Delta I > 0)$	Δ / Π	β_1	β_2
PE	0.527	0.473	-0.098	0.081	-0.013	0.321	0.023
GE	0.918	0.082	-0.169	0.041	-0.152	0.040	0.209

(b) Firm-value effects.

Table 8
The impact of investment flexibility and information allocation on aggregate productivity. The first row of the table below reports aggregate TFP losses computed relative to a model where firms do not face any uncertainty regarding their contemporaneous productivity. The measures are reported as percentage decreases relative to the reference. The second and third rows report the gains to aggregate output relative to the relevant counterfactual.

	Baseline	No Investment Flexibility	Homogeneous Net Benefit
	(1)	(2)	(3)
TFP losses	-0.199	-0.338	-0.239
GDP (FE)	—	0.848	0.240
GDP (GE)	—	0.359	0.102

(c) Aggregate productivity effects.

Read:

- Figure 6 shows that the value of flexibility decreases in productivity, while the value of information increases in productivity.
- Table 7 shows that investment flexibility raises firm value for all firms in partial equilibrium.
- Information reallocation produces gains for high-productivity firms but losses for many low-productivity firms.
- Table 8 shows that removing investment flexibility raises TFP losses from 0.199% to 0.338%.
- Homogeneous information allocation raises TFP losses to 0.239%.

Economic message: Flexibility has the larger aggregate effect, but information allocation matters because better information is most productive when used by high-productivity firms.

7 Short summary

- The paper studies how capital budgeting mitigates uncertainty-driven investment misallocation.
- The data are Japanese firm-level survey data with forecasts, plans, and realizations.
- Firms face substantial uncertainty when setting investment plans.
- Higher-productivity firms face lower uncertainty.
- Investment plans deviate meaningfully from realized investment, implying some flexibility.
- Planned investment is related to expected productivity.
- Realized investment is related to realized productivity even after controlling for plans.
- The model has three stages: information acquisition, plan formation, and actual investment.
- Firms choose signal precision before setting plans.
- Firms can revise investment after productivity is observed, but doing so is costly.
- The marginal benefit of learning is higher for more productive firms.
- Information allocation matters for aggregate TFP because information and productivity are complements.
- Investment flexibility reduces aggregate uncertainty losses by about 41%.
- Information allocation reduces aggregate uncertainty losses by about 17%.
- Flexibility mainly helps lower-productivity firms; information mainly helps higher-productivity firms.

8 Conclusion

8.1 Core conclusion

Charoenwong, Kimura, Kwan, and Tan show that capital budgeting is an economically important mechanism for reducing the impact of uncertainty on investment and productivity. Firms do not simply commit to investment under time-to-build and wait for mistakes to materialize. They acquire information before planning and adjust investment after new information arrives.

8.2 Economic intuition

The intuition is that uncertainty creates investment mistakes. Information acquisition reduces mistakes before the plan is set, while investment flexibility reduces mistakes after the plan is set. These two tools are not equally valuable for all firms. Less productive firms benefit more from flexibility because they rely more on ex-post adjustment. More productive firms benefit more from information because the marginal return to precise forecasting is higher.

8.3 Takeaway

The paper reframes capital budgeting as an intangible technology for managing uncertainty. Its main quantitative message is that state-contingent planning and information acquisition materially reduce capital misallocation, with flexibility providing the larger aggregate gain in the calibration.